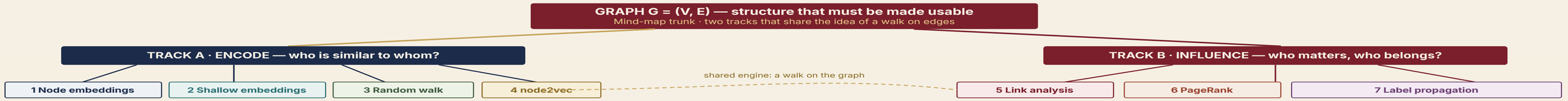
SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING · CONSTITUENT OF SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

STUDENT NAME	PERMANENT REGISTRATION NO. (PRN)	PROGRAMME	CA2 · ONLINE · 9 SEPT 2026
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CA2 · CONCEPT-TO-APPLICATION MIND MAP

From a Graph to a Decision: Embeddings, Walks and Influence

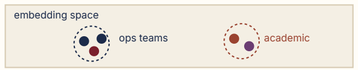
A graph stores who is linked to whom. A model cannot use raw node IDs. This map follows one question through seven tools: how do we turn structure into a vector, a rank, or a label that we can actually act on?



1. Node embeddings CONCEPT

Each node v is mapped to a vector $z_v \in \mathbb{R}^d$ so that geometry in $\mathbb{R}^d$ copies closeness on the graph. Discrete IDs cannot generalise; a vector can be compared, clustered, or scored for a missing edge.

Desired rule: $\text{sim}_G(u, v) \approx z_u^\top z_v$
Softmax context: $P(n \mid v) = \exp(z_n^\top z_v) / \Sigma_u \exp(z_u^\top z_v)$



If two fest teams keep appearing in the same working chains, their vectors sit nearby even when no single edge says “same type of work”.

Application. Recommend a coordination partner for a new intern: nearest neighbour of the PR-cell vector in embedding space, not a manual org-chart lookup.

2. Shallow embeddings METHOD FAMILY

The encoder is only a lookup table. Node v has its own row in $Z \in \mathbb{R}^{|V| \times d}$. There is no filter shared across nodes, so a brand-new vertex has no row until it is retrained. DeepWalk, LINE and node2vec all sit here.

$\text{ENC}(v) = Z_v$ (one row per node)
Reconstruction view: $S \approx \sigma(Z\, Z^\top)$, S = proximity (edge, k-hop, or walk)

Shallow (this CA): lookup row, transductive, DeepWalk / LINE / node2vec.
Not shallow: a GNN shares one filter W on every neighbourhood and can embed a node that was absent at train time.

A shallow table cannot share weights. That is a design choice, not a bug, when the campus roster is fixed for a semester.

Application. Pre-compute one vector per registered society before the sports fest, then query similar societies in milliseconds during stall allocation.

3. Random walk SAMPLER

A walk is a path $v_0, v_1, \ldots$ that hops to a random neighbour. It turns the graph into “sentences” of node IDs. DeepWalk treats those sentences the way skip-gram treats text.

$P(j \mid i) = A_{ij} / d_i$ · undirected occupancy $\pi_i = d_i / 2|E|$



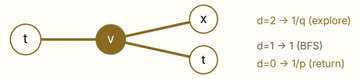
one emitted sentence: H → PR → V

Application. A volunteer “bouncing” between stalls traces the same statistics a walker traces: local traffic, not the full adjacency matrix.

4. node2vec BIASED WALK

Grover & Leskovec (2016). After the step $t \to v$, the next node x is not chosen uniformly. Distance d_{tx} reweights the edge, so the walk can hug a clique or hunt for similar roles.

After $t \to v$: $\alpha_{pq}(t,x) = 1/p$ if $d_{tx}=0$; 1 if $d_{tx}=1$; $1/q$ if $d_{tx}=2$
$\pi_{vx} = \alpha_{pq}(t,x) \cdot w_{vx}$



Small p stays local (homophily). Small q steps outward and groups nodes with the same job in the graph (structural equivalence). Context window then trains skip-gram with negative sampling.

Application. Low q pulls PR, Hospitality and Faculty Advisor together as coordinators even if they do not share a dense clique. High p keeps the cultural stall cluster tight.

5. Link analysis WHO ENDORSES WHOM

Here the question is not “who looks alike” but “who is pointed to”. Web links, paper citations and case citations are directed endorsements. Two classical readings of the same adjacency matrix A:

Prestige (one score): PageRank on the column-stochastic walk.
HITS: $a \propto A^\top h$, $h \propto A\, a$ (normalise each round)



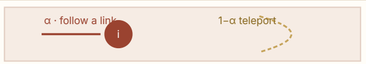
An *authority* is cited by good hubs. A *hub* points to good authorities. On the fest graph, Sports Council behaves like an authority (others depend on its timetable). PR Cell behaves like a hub (it routes people to many teams).

Application. In a judgment-citation network, a frequently cited order is an authority; a review note that cites many orders is a hub. Useful when ranking statutes before a legal-research query.

6. PageRank STATIONARY WALK

Page, Brin et al. A walker follows an out-link with probability α and jumps anywhere with probability $1-\alpha$. The jump (teleport) kills rank sinks and dangling nodes. On an undirected graph the ranking stays close to degree; teleport only softens the gap.

$PR(i) = (1-\alpha)/n + \alpha \Sigma_{j \in B(i)} PR(j) / L(j)$
Matrix form: $r = \alpha P^\top r + (1-\alpha)\, n^{-1} 1$, typically $\alpha = 0.85$



Simple-walk occupancy on the 9-edge fest graph is $\pi_i = d_i/18$. PR Cell and Volunteers both have degree 4, so each holds about 0.22 of the walk mass — the two junctions that must not fail on event day.

Application. Rank campus pages for search; rank committees by information load; rank cited cases so a legal assistant surfaces the judgment that the rest of the corpus actually leans on.

7. Label propagation SPREADING TAGS

A few nodes already know their class. Every other node takes a smoothed average of its neighbours, while labelled nodes are clamped. The harmonic solution of Zhu et al. is the closed form of that smoothing. Raghavan’s majority-vote rule is the unsupervised cousin used for communities.

Iterate: $Y \leftarrow D^{-1} A\, Y$, then reset labelled rows to Y_L.
Harmonic block: $L_{UU} f_U = W_{UL} f_L$

Seed Sports = Operations and Tech = Academic. Hospitality leans Operations (it touches Sports). Faculty Advisor leans Academic (it touches Tech). Volunteers sit on the cut and stay mixed — which is the honest answer, not a failure of the algorithm.

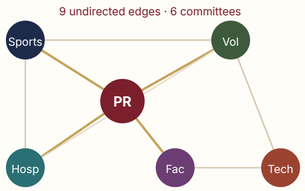
Application. Tag unlabelled students in a club graph from a handful of known members; assign fest duties; fill missing topic labels on a citation graph.

How the seven ideas lock INSIGHT

- A **random walk** is the common physical story. node2vec *samples* it; PageRank *solves* its stationary distribution (with a teleport).
- **Shallow embeddings** store one vector per node. node2vec is the shallow method that feeds walk contexts into skip-gram.
- **Embeddings** answer similarity. **PageRank** answers importance. **Label propagation** answers membership. Link prediction (score $z_u^\top z_v$) sits on the cut between embeddings and link analysis.
- If the roster will grow mid-semester, a lookup table is the wrong encoder; a GNN (out of this CA’s scope, but the contrast matters) shares weights and can embed a new node from its neighbours.

One-line map. Encode with walks → rank with a walk → spread labels on the same edges. Three questions, one graph.

Worked example — SIT Fest Collaboration Graph (original)



Nodes: PR Cell, Sports, Hospitality, Volunteers, Faculty Advisor, Tech Club. Edges = “worked together during the institutional sports fest” (9 links).

- Degrees: PR=4, Vol=4, Sports=3, Hosp=3, Tech=2, Fac=2.
- Walk mass $\pi = d/18$: PR and Vol ≈ 0.22 each. They are the busiest junctions.
- HITS reading: PR is the hub; Sports is an authority for the operations side.
- LPA seeds: Sports = Ops, Tech = Academic. Hosp follows Ops; Fac follows Academic; Vol stays on the cut.

Same graph, seven readings

- **Embeddings:** PR sits near Hosp because they co-occur on walks, even though their “job titles” differ.
- **Shallow table:** six rows in Z; a new stall added on event morning has no vector until retraining.
- **Walk:** H → PR → Vol is a typical sentence DeepWalk would emit.
- **node2vec:** small q groups PR / Hosp / Fac as coordinator roles (structural equivalence).
- **Link analysis:** directed version of “who reports to whom” separates hubs from authorities.
- **PageRank:** protect PR and Volunteers first; they carry the walk.
- **LPA:** do not force Volunteers into one bucket — mixed membership is the signal for floating duty.

Sources used (not copied)

Explanations and the fest graph are written in my own words. Algorithms are the standard ones taught in graph mining; I am attributing them, not quoting textbooks.

- Perozzi et al., DeepWalk (KDD 2014)
- Grover & Leskovec, node2vec (KDD 2016)
- Page, Brin, Motwani, Winograd — PageRank
- Kleinberg — HITS
- Zhu, Ghahramani, Lafferty — harmonic LPA
- Raghavan et al. — community LPA

Originality: definitions, formulae layout, SIT Fest Collaboration Graph, legal-citation application and the two-track mind map are my own academic work for CA2. No paragraph is taken from a website, slide deck or generative write-up.